

Antibody glycoengineering strategies in mammalian cells

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Abstract

As a key parameter impacting functional and structural heterogeneity, protein glycosylation is a critical quality attribute for antibody biotherapeutic manufacturing. The glycan patterns on recombinant antibodies, particularly on the conserved fragment crystallizable (Fc) region, can have significant effects on an antibody's functional activities including clearance rate, antibody-dependent cellular cytotoxicity (ADCC), complement-dependent cytotoxicity (CDC), and anti-inflammatory activity. In this review, we examined specific glycan attachments (fucosylation, sialylation, galactosylation, high-mannose, and bisecting glycans) and their importance to antibody properties. Next, we summarized the recent and current achievements on controlling antibody glycoforms in Chinese hamster ovary (CHO) and other mammalian cells through multiple strategies including genetic engineering, protein engineering, media modification, and other emerging technologies. Further, the impact of one carbohydrate modification on other glycan structures is also described. Finally, approaches to generate desirable homogenous glycan profiles on antibodies are also detailed. By applying multiple complementary intracellular and extracellular strategies, biotechnologists are well on their ways to precisely tuning antibody glycoforms emerging from bioreactors in the coming decades.

KEYWORDS

ADCC, antibody glycosylation, CDC, defucosylation, fragment crystallizable (Fc) region, sialylation

1 | INTRODUCTION

Belonging to the immunoglobulin (Ig) superfamily, antibodies are large, Y-shaped glycoproteins secreted by plasma cells. As adaptor molecules in the immune system, antibodies can recognize and neutralize pathogens from bacteria, viruses, fungi, and other disease-causing agents and induce a defense cascade to facilitate the elimination of pathogens and resolve infections (Buss, Henderson, McFarlane, Shenton, & de Haan, 2012; Jefferis, 2005). In humans, five immunoglobulin classes have been identified (IgG, IgM, IgA, IgE, and IgD), distinguished in their different constant region structures and properties (Figure 1). All immunoglobulins are constructed initially in the same way from paired heavy and light chains (Amzel & Poljak, 1979). In serum, the

most abundant immunoglobulin class with the longest half-life is IgG comprising 75% of antibodies in circulation (Figure 1). IgA and IgM follow with 15% and 10%, respectively, while IgD and IgE are the least abundant in the serum circulation (0.25% and less than 0.01%, respectively) (Figure 1) (Shade & Anthony, 2013).

Human IgGs consist of four isotypes, IgG₁₋₄, that differ in their γ -chain sequences and disulfide bridging patterns, but sharing the same basic structure (Figure 1) (Arnold, Wormald, Sim, Rudd, & Dwek, 2007). The structure of IgG is composed of two light and two heavy chains in covalent and non-covalent association to form three independent protein moieties connected through a flexible linker or hinge region (schematic representation in Figure 2a) (Jefferis, 2005). These moieties are symmetric with two fragment antigen-specific binding (Fab)

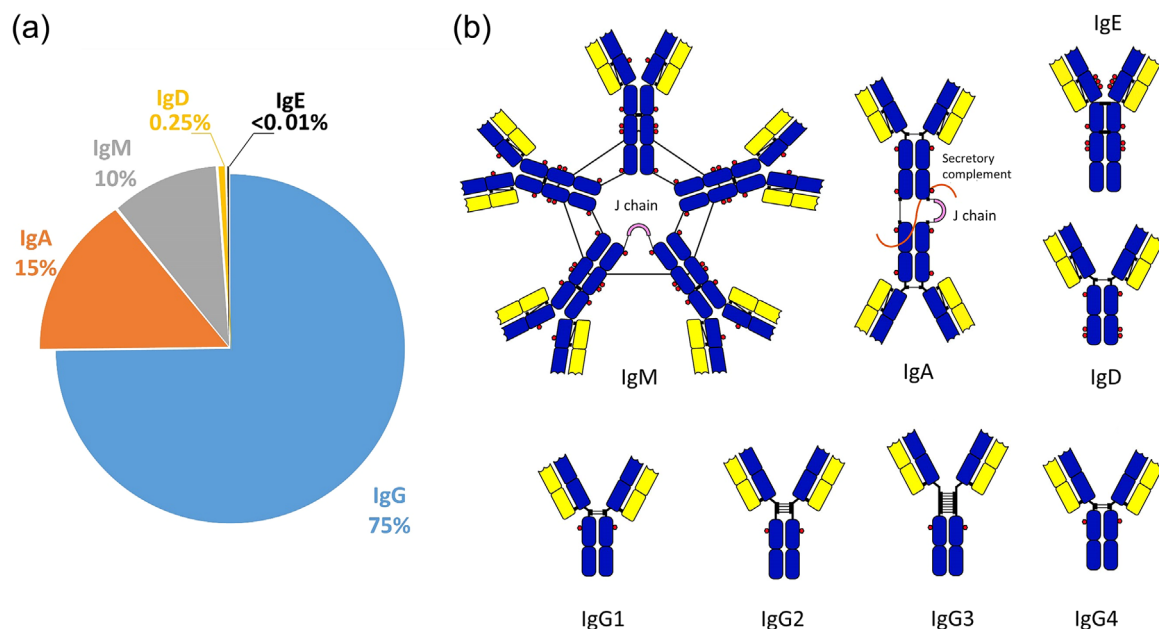


FIGURE 1 Classes of immunoglobulins. (a) The abundance of five immunoglobulins in human blood (Shade & Anthony 2013). (b) The general structures of immunoglobulins isotypes with N-glycans labeled as red hexagons. Four subclasses of IgGs are also presented. Light chains are highlighted in yellow and heavy chains are highlighted in blue

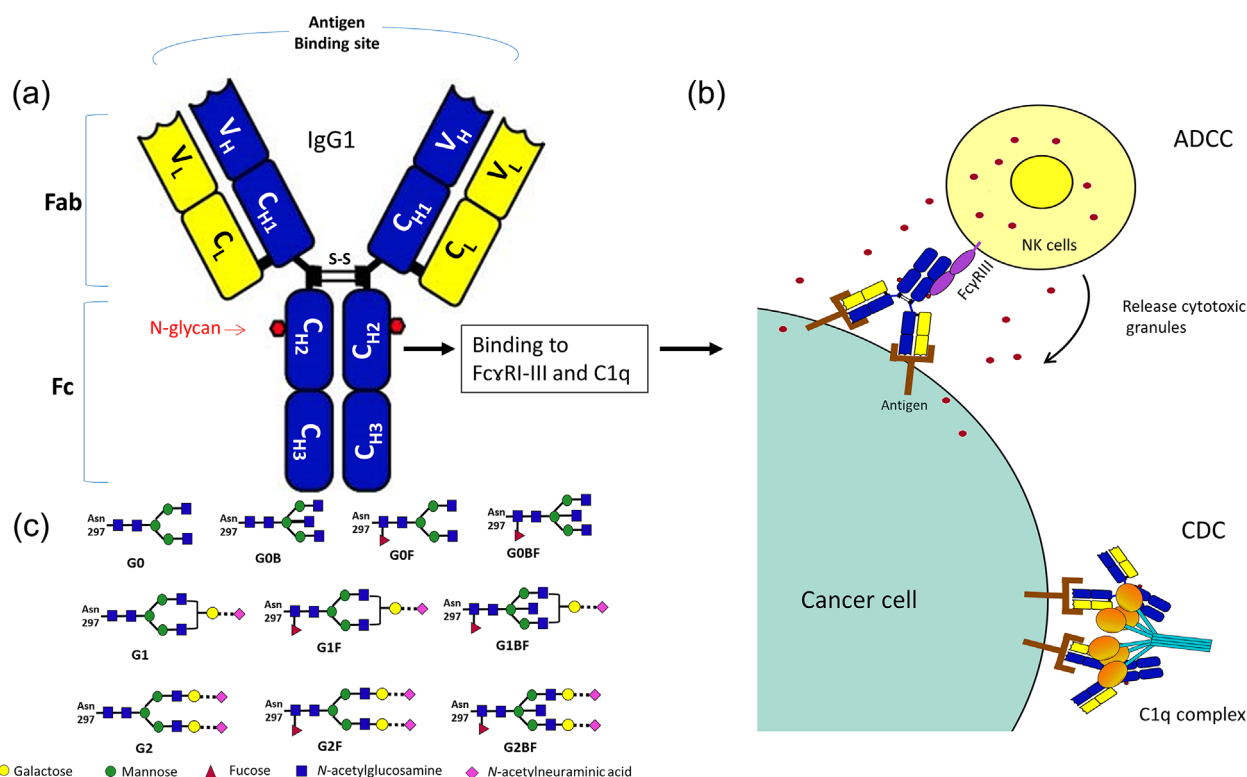


FIGURE 2 The Fc glycans on IgG affect antibody effector functions. (a) Human IgG1 structure with glycans attached at Asn297 N-glycosylation site in the CH2 of the Fc region. C, constant domain; V, variable domain; H, heavy chain; L, light chain; S-S, disulfide bond; Fab, fragment antigen binding domain; Fc, fragment crystallizable domain. Fc region binds effector molecules and cells. (b) The schematic diagram of antibody-dependent cellular cytotoxicity (ADCC) and complement-dependent cytotoxicity (CDC) mechanisms in immunology. (c) The representative glycan structures identified on antibody Fc region

domains and one fragment crystallizable (Fc) domain, containing the interaction sites critical to the effector functions (Buss et al., 2012; Jefferis, 2005). Antibodies bind to specific antigens through the Fab domain and bind to Fc receptors or other proteins on multiple cell types through the Fc domain to initiate distinct effector mechanisms (Liu, 2015).

The numerous applications of recombinant monoclonal antibody (rMAb) therapeutics ranging from diagnosis to treatment of major diseases including cancers, autoimmune disease, and inflammatory disorders have continuously expanded, in step with technological improvements and innovations in the development and production of rMAbs (Niwa & Satoh, 2015). To date, the majority of recombinant glycoproteins in production from the most widely used mammalian cell lines such as Chinese hamster ovary (CHO), mouse myeloma (NS0) and (Sp2/O) are recombinant monoclonal antibodies, including chimeric, humanized, or human immunoglobulin Gs (IgGs) (Raju & Jordan, 2012).

One critical post-translational modification of rMAbs is glycosylation, particularly in the Fc region of the IgG. Although oligosaccharides represent only 3% of the total mass of the IgG, the particular glycoforms are involved in many essential immune effector functions, such as inducing the anti-inflammatory response, altering the circulatory half-life, killing antibody-coated pathogens by antibody-dependent cellular cytotoxicity (ADCC), and complement-dependent cytotoxicity as shown in Figure 2b (Jefferis, 2009; Jennewein & Alter). ADCC is the mechanism that triggers the apoptosis of antibody-coated targeted cells by natural killer (NK) and other effector cells in which the FcγRIII receptor on their surfaces recognize the Fc region of the bound antibody. Once activated, NK cells will release cytotoxic granules inducing apoptosis of the target cells. Another antibody-induced pathway is the classical pathway in the complement system, or often called complement-dependent cytotoxicity (CDC). CDC is initiated when C1q complex interacts with antibodies bound to the pathogen surface and then facilitates the opsonization and phagocytosis of pathogens by forming the membrane attack complex (MAC) to induce osmotic lysis of pathogen cells (Beck, Wurch, Bailly, & Corvaia, 2010; Jefferis, 2005). The carbohydrate moieties are also helpful in maintaining an open conformation of the two heavy chains of IgG and preserving the structural integrity of the FcγR-binding site (Yamaguchi et al., 2006). Effector mechanisms mediated through FcγRs, C1q, mannose-binding lectin (MBL), and mannose receptor (MR) are severely compromised for aglycosylated IgG-Fc (Jefferis, 2005). Deglycosylated IgG antibodies are unable to mediate an *in vivo* triggered inflammatory response (Beck, Wagner-Rousset, Ayoub, Van Dorsselaer, & Sanglier-Cianferani, 2013; Nimmerjahn & Ravetch, 2006).

Unlike IgA, IgD, IgM, and IgE that have several sugar moieties attached to their Fc region as shown in Figure 1, the IgG isotypes are characterized only by one sugar moiety attached to the Asparagine 297 in each of the CH2 domains of Fc region (Figure 2a) (Bohm, Schwab, Lux, & Nimmerjahn, 2012). Most human serum IgGs, possess primarily bi-antennary glycoforms with a trimannosyl core decorated with a range of additional sugars including fucose residues attached at the innermost N-acetylglucosamine (GlcNAc) and branches containing

GlcNAc, galactose, and N-acetylneuraminic acid (Neu5Ac) residues (see Figure 2c). The Fc N-glycans lie within a pocket formed by the 2 CH2 domains of the antibody where they interact with internal amino acid residues (Raymond et al., 2015; Wormald et al., 1997). As a consequence of this embedment, the Fc N-glycans are mostly limited to bi-antennary structures with partial galactosylation and low sialylation (Jefferis, 2005; Raymond et al., 2015). The majority of human IgG Fc glycans are highly fucosylated (>80%) with 0 (G0F ~35%), 1 (G1F ~35%), and 2 (G2F ~16%) galactose residues (Arnold et al., 2007; Wuhrer et al., 2007). A small proportion of human IgG Fc glycans contain a bisecting GlcNAc (G1BF and G2BF in Figure 2c) (Wuhrer et al., 2007). Only 5–10% of human IgG Fc glycoforms are mono-sialylated and less than 5% of IgG Fc glycoforms possesses two sialic acids (Arnold et al., 2007; Shade & Anthony, 2013; Stadlmann et al., 2009; Wuhrer et al., 2007; Zauner et al., 2013). Recombinant IgGs expressed in CHO, mouse myeloma NS0, and Sp2/O cells are in general also highly fucosylated with low level of galactosylation and sialylation (Raju & Jordan, 2012). A representative glycan profile from IgG1 expressed in CHO-K1 cells using MALDI-TOF mass spectrometry analysis is presented in Figure 3 (Chung, Wang, Yang, Yin, et al., 2017). Specific glycoforms on a rMAb can have significant effects on the drug's bioactivities and therapeutic properties, such as high-mannose glycans can increase antibody clearance rates (Yu et al., 2012), afucosylation, bisecting GlcNAc residues, and high-mannose glycans can enhance ADCC activity (Jefferis, 2009; Shi & Goudar, 2014; Zhong et al., 2012) and galactosylation can improve CDC activity (Liu, 2015; Peschke, Keller, Weber, Quast, & Lünemann, 2017).

Given the critical roles of Fc N-glycans on antibody activity, we will focus this review on the recent and current glycoengineering efforts aimed at altering fucosylation, presence and absence of GlcNAc attachments, galactosylation, sialylation, and the generation of high-mannose glycans. Efforts to further tailor rMAbs glycoforms, including reducing heterogeneity of glycans and emerging technologies for producing antibodies bearing homogenous glycan profile are also described.

2 | ANTIBODY GLYCOENGINEERING STRATEGIES

2.1 | Defucosylation

Numerous studies have shown that the core-fucose residue of the CH2-associated N-linked oligosaccharides plays a critical role in regulating the magnitude of antibody effector function ADCC (Iida et al., 2006; Satoh, Iida, & Shitara, 2006; Shields et al., 2002; Shinkawa et al., 2003; Yamane-Ohnuki et al., 2004). Compared to the fucosylated antibodies from wild-type CHO cells, defucosylated antibodies exhibited 100-folds higher ADCC *in vitro* (Peipp et al., 2008). The core fucose is bound through a α -1,6 linkage to the innermost GlcNAc residue on an N-oligosaccharide (Figure 4), which can hinder Fc binding to Fc receptors of lymphocytes (Yamane-Ohnuki et al., 2004). It has been demonstrated that the antibody ADCC enhancement due to fucose elimination is

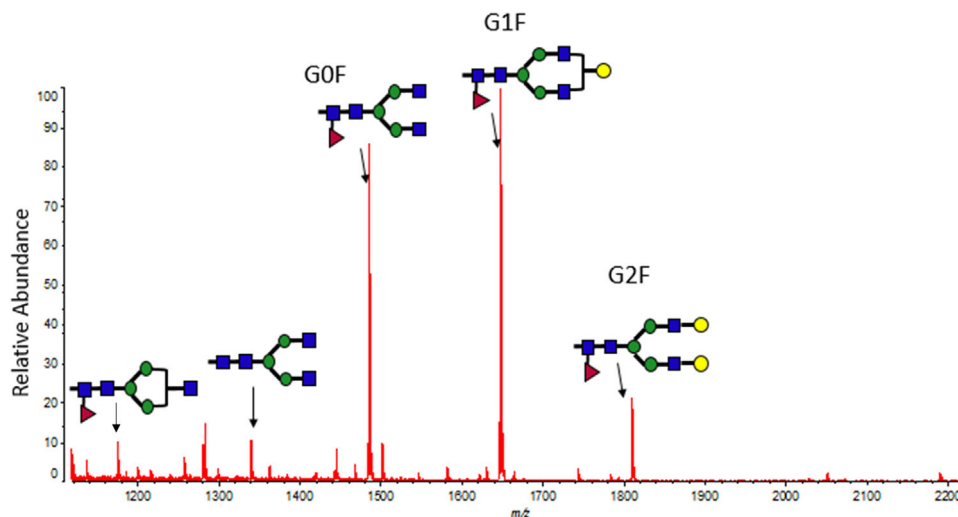


FIGURE 3 MALDI-TOF mass spectrometry glycan analysis of IgG1 expressed in CHO-K1 cells

attributed to a subtle conformational change in a limited region of IgG1-Fc, resulting in higher Fc binding to the Fc-gamma receptor IIIa (FcγRIIIa) on NK cells (Iida et al., 2006; Kanda, Yamada et al., 2007; Matsumiya et al., 2007).

Moreover, antibodies with Fc regions containing nonfucosylated N-glycans exhibit dramatically enhanced ADCC without any detectable change in CDC or antigen binding capability (Yamane-Ohnuki et al., 2004; Yamane-Ohnuki & Satoh, 2009). The lack of fucose is primarily responsible for controlling binding to FcγRIIIa receptor and in the recruitment of mononuclear cells (primarily NK cells) as effector cells for ADCC (Okazaki et al., 2004; Peipp et al., 2008; Shields et al., 2002). However, studies on polymorphonuclear cells (PMNs) found that highly fucosylated antibodies were preferentially more effective in triggering PMN, which mediated tumor cell lysis via Fc gamma RII receptor (FcγRII) (Peipp et al., 2008). Thus, the impact of Fc fucosylation still needs further exploration and may be critically dependent on the specific effector cell type recruited. Given that the majority of monoclonal antibodies on the market are targeted to mediate cytotoxicity of NK effector cells, we will focus on the defucosylation of antibodies here.

2.1.1 | Inactivation of α1,6 fucosyltransferase—Disruption of FUT8 gene

Inactivation of the α1,6 fucosyltransferase has been extensively studied and still represents the most efficient cellular modification preventing the addition of fucose to the core of the complex N-glycan. In mammals, the FUT8 gene encodes α-1,6 fucosyltransferase in the medial Golgi cisternae that catalyzes the transfer of fucose residues from GDP-fucose to the innermost GlcNAc of the tri-mannosyl core structure via the α1,6 linkage (Miyoshi et al., 1999). In order to generate non-fucosylated antibodies, disruption of the FUT8 gene has been investigated via various approaches as discussed below (Ma, Simala-Grant, & Taylor, 2006; Mori et al., 2007; Mori et al., 2004; Yamane-Ohnuki et al., 2004).

Traditionally, numerous methods such as antisense cDNA, shRNA, or transgenic overexpression were used to reduce or remove fucosylation (Stanley, 2014). The introduction of small interfering RNAs (siRNAs) against the FUT8 gene reduced FUT8 mRNA expression to 20% in CHO/DG44 cells and combined with *Lens culinaris agglutinin* (LCA) selection yielded 60% antibody defucosylation with over 100-folds higher ADCC (Mori et al., 2004). Disrupting both FUT8 alleles in CHO/DG44 cell line by sequential homologous recombination produced completely defucosylated recombinant antibodies (Yamane-Ohnuki et al., 2004). Chimeric anti-CD20 IgG1 produced from FUT8^{-/-} CHO cells showed the same level of antigen-binding activity and CDC as the FUT8^{+/+} produced (Yamane-Ohnuki et al., 2004). However, the homologous recombination strategy is time-consuming and laborious to produce completely nonfucosylated antibodies or the complete knockout of FUT8 is reversible and the difficulty of recapitulating the knockout in protein-production cell lines has prevented the widespread adoption of homologous recombination FUT8^{-/-} cells as hosts for antibody production (Malphettes et al., 2010; Stanley, 2014; Yamane-Ohnuki et al., 2004). To improve the efficiency of disrupting FUT8 gene, technologies such as zinc-finger nucleases (ZFNs) and transcription activator-like effector nucleases (TALENs) were developed and reported (Cristea et al., 2013; Malphettes et al., 2010). Using ZFN to disrupt the FUT8 gene in a region encoding the catalytic core of the enzyme produced FUT8^{-/-} CHO cells in 3 weeks at a frequency of 5% in the absence of any selection, and these lines produced antibodies completely lacking core fucosylation but having an otherwise normal glycosylation pattern without adverse phenotypic effects (Malphettes et al., 2010). Separately, Cristea et al. (2013) demonstrated knockout of FUT8 gene via simultaneous TALE nuclease-mediate integration of an antibody cassette. Composed of sequence-specific DNA-binding domains connected to a nonspecific DNA nuclease, ZFNs, and TALENs enable a broad range of genetic modulations by inducing DNA double-strand breaks that stimulate error-prone nonhomologous end joining (NHEJ) or homology-directed repair (HDR) at specific genomic

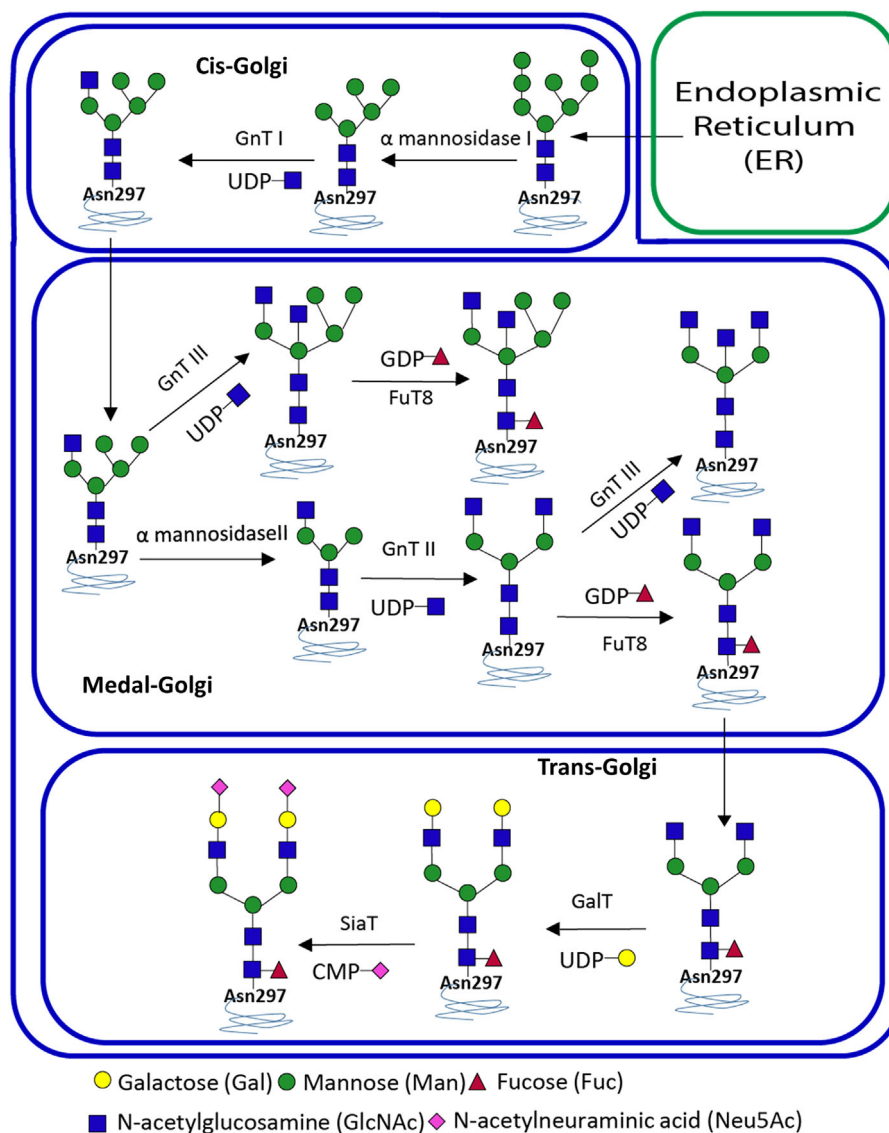


FIGURE 4 A simplified antibody N-glycosylation pathway in mammalian cells. After translocation from the ER to the cis-Golgi, the $\text{Man}_8\text{GlcNAc}_2$ glycoform is trimmed by Golgi α -mannosidases I to generate high mannose glycans including $\text{Man}_5\text{GlcNAc}_2$. Then an N-acetylglucosaminyltransferase (GnT-1 or Mgat1) adds a GlcNAc to $\text{Man}_5\text{GlcNAc}_2$ to yield $\text{GlcNAcMan}_5\text{GlcNAc}_2$. Hybrid N-glycans are generated when the $\text{GlcNAcMan}_5\text{GlcNAc}_2$ structure exposes two terminal Man residues. The majority of N-glycans are tailored by Golgi α -mannosidase II to remove the two terminal mannoses and generate $\text{GlcNAcMan}_3\text{GlcNAc}_2$. Then the enzyme β -1,2-N-acetylglucosaminyltransferase II (GnT-II or Mgat2) adds a second GlcNAc to this $\text{GlcNAcMan}_3\text{GlcNAc}_2$ structure to generate the glycan product $\text{GlcNAc}_2\text{Man}_3\text{GlcNAc}_2$, which is the precursor for complex N-glycans. Additionally, another GlcNAc can be added to the innermost Man group by the enzyme β 1,4-N-acetylglucosaminyltransferase III (GnT-III or Mgat3) in the medial Golgi, resulting in bisecting GlcNAc structures. Additional modifications of complex and hybrid N-glycans can occur in the Golgi and include the addition of α 1,6-fucose to the innermost GlcNAc adjacent to Asn by α 1,6-fucosyltransferase. The N-glycans can be extended by adding a β -linked galactose residue to GlcNAc to produce $\text{Gal}\beta$ 1-4 GlcNAc and terminally sialylated by different sialyltransferases

locations (Gaj, Gersbach, & Barbas, 2013; Urnov, Rebar, Holmes, Zhang, & Gregory, 2010). Thus, NHEJ can be harnessed to mediate functional disruptions of target genes, as indels occurring in the coding region could lead to frameshift mutations and protein function loss (Perez et al., 2008; Sun et al., 2015).

More recently, distinct from the protein-based ZFNs and TALENs, the CRISPR (clustered regularly interspaced short palindromic repeat)/Cas9 (CRISPR-associated protein 9) system, derived from the microbial

adaptive immune system (Barrangou et al., 2007), has emerged as an efficient alternative for inducing targeted genetic alterations (Gaj et al., 2013), and applied to various species such as plants, animals, bacteria, and yeast (DiCarlo et al., 2013), as well as CHO cells (Ronda et al., 2014; Sun et al., 2015). Instead of protein-DNA binding as used in ZFN and TALEN systems, the CRISPR system requires only Cas9 nuclease and single guide RNA (gRNA) binding to targeted DNA sequences (Mali et al., 2013), providing several advantages including ease of

customization, higher knockout efficiency, and multiplexed genome editing capability (Ran et al., 2013). Sun et al. (2005) reported functional FUT8 disruptive clones (FUT8^{-/-}) in CHO cells using CRISPR/Cas9 were generated within 3 weeks at indel frequencies ranging from 9% to 25%, which was further enhanced to 52% with LCA selection. The derived FUT8^{-/-} clone was able to produce defucosylated therapeutic mAb with no detrimental effects on cell growth, viability, or product quality (Sun et al., 2015). Moreover, Grav et al. (2015) employed CRISPR/Cas 9-targeted FUT8, BAK, and BAX triple knockouts simultaneously and created a cell line producing afucosylated antibody with improved protein productivity.

2.1.2 | Overexpression of GnT-III

A different and complementary approach to improve an antibody's biological activity would be to overexpress β -1,4-N-acetylglucosaminyltransferase III (GnT-III) enzyme to produce antibodies enriched in bisecting oligosaccharides. GnT-III is a Golgi-localized enzyme that catalyzes the addition of a GlcNAc residue onto a bisecting position of the N-linked oligosaccharide, as depicted in Figure 4 (Narasimhan, 1982). Overexpression of the GnT-III gene in a CHO cell line expressing an anti-neuroblastoma IgG1 resulted in enhanced ADCC activity because bisecting GlcNAc inhibits fucosylation and the conversion of hybrid to complex glycan in the N-glycan biosynthetic pathway, perhaps due to the steric hindrance of GlcNAc-Man-Man structure by the bisecting GlcNAc (Ferrara, Brunker et al., 2006; Schachter, 1986; Umana, Jean-Mairet, Moudry, Amstutz, & Bailey, 1999). Thus, GnT-III overexpression leads to an increase in non-fucosylated and hybrid oligosaccharides content, which further enhances binding affinity for Fc γ Rs (Ferrara, Brunker et al., 2006; Ferrara, Stuart, Sondermann, Brunker, & Umana, 2006). Overexpression of the GnT-III in a CHO cell line producing anti CD-20 antibody resulted in a 20-folds lower antibody dosage in order to achieve the same ADCC effect induced by the antibody produced from the parental CHO cell line (Davies et al., 2001). Furthermore, by fusing the catalytic domain of GnT-III with the localization domain (cytoplasmic, transmembrane, and stem region) of other Golgi-resident enzymes in the N-glycosylation pathway of HEK293-EBNA cells, a chimeric GnT-III fusion protein was expressed that can compete even more efficiently against the endogenous core α 1,6-fucosyltransferase, yielding principally bisected non-fucosylated hybrid-rich glycans (Ferrara, Brunker et al., 2006). Golgi α -mannosidase II (Man II) yields the GlcNAc₂Man₃GlcNAc₁ precursor for all biantennary, complex N-glycans as illustrated in Figure 4. Co-expressing both GnT-III and Man II led to a higher proportion of bisected non-fucosylated complex-rich glycans (Ferrara, Brunker et al., 2006). While both glycovariants featured strongly increased ADCC activity compared to the wild-type antibody, the hybrid-rich antibody showed reduced CDC compared to complex-rich or wild-type counterparts (Ferrara, Stuart et al., 2006; Schuster et al., 2005). It is worth noting that the effect of bisecting GlcNAc of the IgG1 on enhancing ADCC is modest when compared to that of defucosylated IgG1, therefore, the lack of fucosylation appears to play the critical

role in the enhancement of ADCC (Davies et al., 2001; Shinkawa et al., 2003; Umana et al., 1999).

2.1.3 | Inhibition of the supply of nucleotide sugar substrate for fucosylation

An alternative approach for producing fucose-negative antibodies is to knockout enzymes responsible for biosynthesis of GDP-fucose, the donor for fucosyltransferase in the Golgi apparatus. The advantage of this approach is that it targets all fucosyltransferases, not just FUT8 (Stanley, 2014). In mammalian cells, GDP-fucose is synthesized by two distinct pathways as illustrated in Figure 5. The *de novo* pathway is characterized by the conversion of GDP-mannose to GDP-4-keto-6-deoxymannose by GDP-mannose 4,6-dehydratase (GMD). Then this keto intermediate is converted to GDP-fucose by GDP-4-keto-6-deoxymannose-3,5-epimerase-4-reductase (GMER or FX protein) (Becker & Lowe, 2003). The *salvage* pathway directly utilizes fucose that is either transported into the cytosol from an extracellular origin or liberated from catabolism of fucosylated glycans in the lysosome and then transported into the cytosol (Becker & Lowe, 2003). The *salvage* pathway is enabled by fucokinase and GDP-fucose pyrophosphorylase (GFPP) (Becker & Lowe, 2003). Subsequently, GDP-fucose synthesized by these two pathways is transported into the lumen of the Golgi apparatus by GDP-fucose transporter (GFT) (Omasa et al., 2008).

Reduction-of-function analyses using siRNA revealed three key genes involved in oligosaccharide fucosylation, including FUT8, GMD, and GFT. Single-gene knockdown of the individual targets was insufficient to completely defucosylate the products in antibody-producing cell lines (Imai-Nishiya et al., 2007). For GFT, the introduction of a CHO GFT siRNA expression plasmid into a human antithrombin III (AT-III)-producing cell line showed approximately 75% reduction of Golgi GDP-fucose, resulting in 10–40% defucosylation in AT-III (Omasa et al., 2008). Similarly, Kanda, Imai-Nishiya et al. (2007) reported expression levels of GFT mRNA could be reduced more than 90% in GFT siRNA-expressing antibody-producing CHO cells, but only a 20–40% reduction of the recombinant IgG Fc oligosaccharide fucosylation was observed. In all, a positive relation between the Fc oligosaccharide fucosylation and the mRNA expression for both GMD and FUT8, but not for GFT, was observed suggesting that GMD and FUT8 are independent bottlenecks for α 1,6 fucosylation (Kanda, Imai-Nishiya et al., 2007).

Alternatively, a mutant of the Pro-CHO 5 cell, Lec 13, has been isolated using LCA selection, with a loss of GMD activity resulting in only 10% of fucose residing on the glycan chain compared to the parental cell line (Ripka, Adamany, & Stanley, 1986). But Lec13 is unsuitable for the serum-free fed-batch production of fucose-negative antibody because the remaining GMD activity in Lec13 accumulates intracellular GDP-fucose via the *de novo* pathway and gradually increases the ratio of fucosylated products with extended culture (Kanda et al., 2006). Additionally, a GMD knockout in CHO/DG44 cells was devoid of intracellular GDP-fucose, but the fucosylation could be

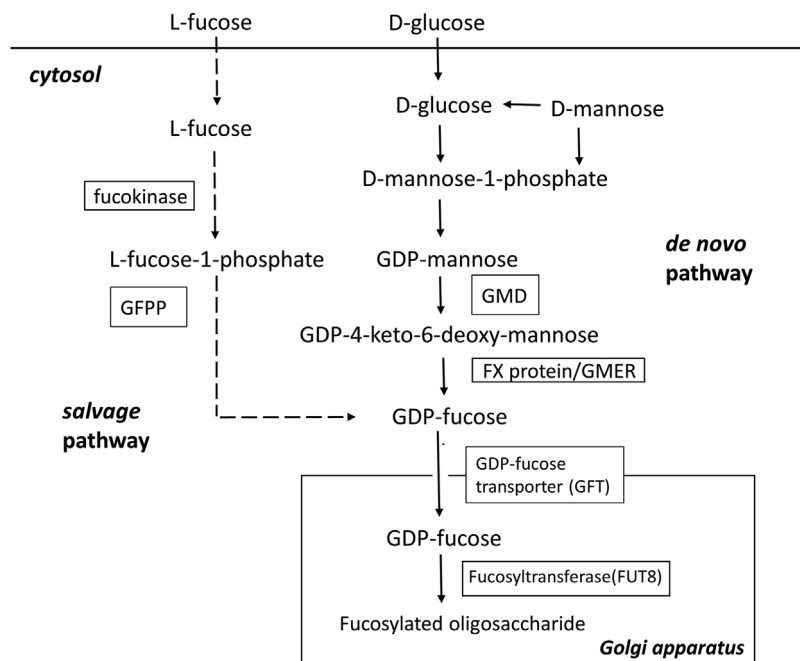


FIGURE 5 The pathways of oligosaccharide fucosylation in mammalian cell lines, modified from Kanda, Imai-Nishiya et al. (2007), GMD, GDP-mannose 4,6-dehydratase; GFPP, GDP-fucose pyrophosphorylase; FX protein/GMER, GDP-4-keto-6-deoxymannose-3,5-epimerase-4-reductase

recovered through the *salvage* pathway upon the addition of *L*-fucose into the culture medium (Kanda, Imai-Nishiya et al., 2007). Besides, a double knockdown of FUT8 and GMD using a short hairpin siRNA tandem expression vector could rapidly convert antibody-producing CHO cells to fully non-fucosylated antibody producers in serum-free fed-batch culture (Imai-Nishiya et al., 2007).

In addition, a consideration of fucosylated and non-fucosylated antibody levels is important in order to control the product's quality and efficacy, especially around ADCC, as required by regulatory authorities (Louie et al., 2016). While a commonly used strategy in CHO host cells to generate defucosylated antibodies is to knock out the FUT8 gene, this approach has a major drawback in requiring parallel cell line development for both the afucosylated and unmodified host cell lines. Even more challenging is large-scale clone screening to obtain primarily unmodified and FUT8-KO clones with highly similar product quality attributes to ensure that any advantage observed on ADCC can be strictly attributed to afucosylation (Louie et al., 2016). To circumvent these issues, other approaches have been explored and developed to evaluate the importance of fucosylation without knocking out or down the FUT8 gene. Given the nature of the fucosylation mechanism in CHO (Figure 5), cells inhibited for the *de novo* pathway will depend on the *salvage* pathway to synthesize GDP-fucose which requires *L*-fucose feeding to the cell culture media. Thus, once the *de novo* pathway is blocked, *L*-fucose can be added as a switch to change antibody glycan profiles from afucosylated to fucosylated glycan variants without affecting other product quality features. FX protein (also called GMER) knock out clones via CRISPR approach can express antibodies with comparable productivity as parental host with a desired ratio of primarily fucosylated to afucosylated glycans when *L*-fucose is titrated into the

production media in order to achieve intended levels of FcγRIII-binding and ADCC for an antibody (Louie et al., 2016).

The use of siRNA successfully knocked down the expression of FUT8, GMD, and GFT in the fucosylation pathway and reduced antibody fucosylation content (Imai-Nishiya et al., 2007; Mori et al., 2004). However, these modifications may cause unpredictable off-target effects that are impractical to implement at the manufacturing level. Likewise, overexpression of the prokaryotic GDP-6-deoxy-D-lyxo-4-hexulose reductase (RMD) enzyme (von Horsten et al., 2010) is another approach to reduce fucosylation by converting GDP-mannose to GDP-6-deoxy-d-lyxo-4-hexulose and reducing synthesis of the GDP-6-keto-6-deoxy-mannose intermediate in the *de novo* pathway (von Horsten et al., 2010). However, complete elimination of fucosylation is a challenge (Kelly et al., 2018). Addition of chemical inhibitors such as 2-deoxy-2-fluoro-*L*-fucose (Winterbourne, Butchard, & Kent, 1979) will also not typically achieve 100% defucosylation. Media supplementation of *D*-arabinose (a close chemical analog of *L*-fucose) was shown to reduce high-mannose N-glycans and provide nearly 100% replacement of fucosylation with arabinosylation (Hossler et al., 2017). Alternatively, the overexpression of GnTIII can generate a heterogeneous glycan population not commonly present on antibodies and may not be desirable for product quality consistency (Louie et al., 2016). Therefore, there are a variety of different cell engineering and media modification strategies that can be applied to alter the fucosylation status and potential ADCC efficacy of antibody biotherapeutics. Choosing which one to use will depend on the effectiveness and practical implementation of the technique as well as a consideration of other factors including biomanufacturing compatibility and availability and intellectual patent considerations.

2.2 | Sialylation

For many therapeutic glycoproteins, the terminal sialic acids on glycans can have a clear and critical role for a protein's circulatory lifespan and efficacy. Serving as a biological mask, the distal sialic acid can shield Gal residues that when exposed prompt the rapid removal of the protein from blood circulation due to endocytosis-mediated uptake by asialoglycoprotein receptors on hepatocytes (Ashwell and Harford, 1982; Cole, Nichols, Poisson, Harnois, & Livingston, 1993; Wang, Stuczynski, Gao, & Betenbaugh, 2015). Therefore, in mammalian cells, it is often highly desirable to maximize the terminal sialic acid content of non-antibody glycoproteins in order to ensure their quality and half-life as an effective therapeutic (Zhang, Koskie, Ross, Kayser, & Caple, 2010). However, due to conformational restrictions, antibody Fc regions that are undersialylated may not be accessible to asialoglycoprotein receptors as is observed for other glycoproteins (Reusch & Tejada, 2015). Alternatively, the half-life of many antibodies is regulated through antibody Fc domain binding to the neonatal FcR (FcRn) through transcytosis at low pH conditions followed by release into the bloodstream at physiological pH prior to lysosomal degradation (Reusch & Tejada, 2015). Even for antibodies, the sialic acid can act to alter other properties of the resulting protein (Reusch & Tejada, 2015), and this topic has gained intense study in recent years. Researchers observed that increasing IgG Fc α 2,6 sialylation enhances the anti-inflammatory activity of intravenous immunoglobulin (Kaneko, Nimmerjahn, & Ravetch, 2006; Lin, Tsai et al., 2015) and sialylation enhancement deteriorates CDC activity (Quast et al., 2015). Decreasing Fc sialylation may affect Fc γ RIII receptor binding onto NK cells followed by an effect on ADCC activity (Naso, Tam, Scallion, & Raju, 2010; Raymond et al., 2015), and lower affinity binding to insoluble or cell-surface antigens, but not affect the antibody binding to soluble antigens (Scallion, Tam, McCarthy, Cal, & Raju, 2007).

2.2.1 | Increase antibody α 2,6 sialylation for promoting anti-inflammatory activity

Intravenous immunoglobulin (IVIg) predominantly consists of pooled polyclonal IgG from thousands of donors and has been used therapeutically to treat inflammatory diseases at high doses (Dwyer, 1992; von Gunten et al., 2014). The sialylation on the glycan of the Fc portion of IVIg has been reported to exhibit superior anti-inflammatory efficacies over desialylated IVIg and deglycosylated IVIg (Kaneko et al., 2006). Both desialylated IVIg and deglycosylated IVIg abrogated the anti-inflammatory effect in vivo (Kaneko et al., 2006). The demand for high doses (1–2 g/kg bodyweight) of unfractionated IVIg is due to a generally low concentration of sialylated IgG in human plasma. In a mice model of arthritis, equivalent levels of anti-inflammatory activity were exhibited when using 1 g/kg IVIg and 0.1 g/kg sialic acid-enriched IVIg, indicating a 10-folds enhancement with the addition of terminal sialic acids (Anthony et al., 2008; Kaneko et al., 2006). Several independent in vivo models further confirmed this theory that the terminal sialylation of Fc glycans is essential to elicit a sufficient anti-inflammatory activity for IVIg (Schwab, Biburger, Kronke, Schett, &

Nimmerjahn, 2012; Schwab et al., 2014). In addition, the presence of α 2,6-linked sialic acid residues, rather than α 2,3-sialic acids, plays a critical role in inducing IVIg-elicited anti-inflammatory activity (Anthony et al., 2008; Anthony & Ravetch, 2010; Scallion et al., 2007).

The major issue in using IVIg to treat auto-immune diseases is the inconsistent therapeutic efficacy and potency at high-dose administration and the lack of sialylated products for clinical use (Washburn et al., 2015). Lectin chromatography using *sambucus nigra agglutinin* (SNA) was considered for enriching sialylated IgG glycovariants in the IVIg preparation, due to this lectin's binding specificity to α 2,6 sialic acids. One analysis of fractions obtained by lectin chromatography revealed that SNA preferentially binds to glycans of Fab, which contains 46.2% sialylated N-glycans, accounting for 15–25% of serum IgG (Bondt et al., 2014; Holland et al., 2006; Jefferis, 2007; Kasermann et al., 2012; Stadlmann, Pabst, & Altmann, 2010), and easier access to sialic acid residues than Fc glycans for SNA lectin (Stadlmann et al., 2009). Besides, experiments with a monoclonal antibody from a human cell line and IVIg Fc fragments indicated that at least two sialic acids in the Fc region of an antibody are required for SNA lectin binding (Stadlmann et al., 2009, 2010). Such glycoforms contain either two monosialylated glycans (S1-S1 glycoform) or a disialylated glycan (S2-S0 glycoform), which represents 1% or less of the total human IgG (Bohm et al., 2012; Stadlmann et al., 2009, 2010).

These restrictions have stimulated researchers and engineers to generate therapeutic alternatives that could leverage the broad mechanisms of action of IVIg while improving therapeutic consistency and potency in a scalable and controllable antibody production process. Washburn et al. (2015) proposed a controlled and scalable process to produce fully hyper-sialylated IVIg (S2-S2 glycoform) by treating native IVIg with in vitro β 1,4-galactosyltransferase and α 2,6-sialyltransferase. The in vivo characterization of this drug candidate revealed consistent and enhanced anti-inflammatory activity up to 10-folds higher than native IVIg across different animal models (Washburn et al., 2015). A mass spectrometric quantification revealed that the IVIg preparation consist primarily of the IgG1 isotype (Stadlmann et al., 2009). Furthermore, a completely α 2,6-sialylated recombinant IgG1 greatly enhanced potency at a much reduced dose of 0.03 g/kg bodyweight compared to 1–2 g/kg bodyweight for native IVIg (Anthony et al., 2008).

In terms of increasing antibody sialylation in cells, the sialylation is controlled primarily by the interaction of local IgG structure with the carbohydrate and the accessibility of the sialyltransferase to sialylate the N-glycan on the polypeptide (Lund, Takahashi, Pound, Goodall, & Jefferis, 1996). Overexpression of α 2,6 sialyltransferase (α 2,6-ST) in a recombinant IgG-producing CHO cell line has been examined for increasing the sialylation content of recombinant IgG (Jassal et al., 2001; Lin, Mascarenhas et al., 2015). For example, transfection of rat α 2,6 sialyltransferase increased the sialylation content in a CHO IgG3 expressing cell line (Jassal et al., 2001). The recent report on the genomic sequence of the CHO-K1 cell line revealed that beta-galactoside α -2,6-sialyltransferase 1 (ST6GAL1) gene is retained in CHO genome, but its expression is severely suppressed and regulated (Xu et al., 2011). The introduction of an endogenous CHO ST6GAL1

gene into the CHO cell line can serve to enhance the α 2,6 sialylation level of an antibody in CHO (Onitsuka et al., 2012). Most attempts to overexpress α -2,6-sialyltransferase in CHO revealed that this approach could only moderately enhance α 2,6-sialylated glycans, and the mixture of α 2,6 and α 2,3 sialylation existed in the final glycoforms (Jassal et al., 2001; Lin, Mascarenhas et al., 2015). Meanwhile, co-overexpression of β 1,4-galactosyltransferase (β 1,4-GalT) and α -2,6-sialyltransferase was implemented to achieve a glycoprofile dominated by α 2,6-sialylation for recombinant antibody production (Raymond et al., 2015). More recently, our group disrupted two α 2,3 sialyltransferases (ST3GAL4 and ST3GAL6) and combined this cell line variant with α 2,6 sialyltransferase overexpression in pursuit of an exclusive α 2,6 sialylated CHO cell line (Chung, Wang, Yang, Yin, et al., 2017). Antibody produced in our α 2,3 sialyltransferase knockout- α 2,6 sialyltransferase overexpressing cells contains solely α 2,6 sialylation (Chung, Wang, Yang, Yin, et al., 2017).

Amino acid substitution of specific residues in the CH2 domain of the IgG molecule provides an alternate route to modulate IgG sialylation level independent of the sialyltransferase overexpression approach discussed above. Replacing Phe²⁴¹ (F241), Phe²⁴³ (F243), Val²⁶⁴ (V264), Asp²⁶⁵ (D265), or Arg³⁰¹ (R301) with alanine can increase sialylation and galactosylation on an IgG3 antibody for each amino acid. In particular, 73% of oligosaccharide chains were monosialylated or disialylated in the F243A mutant and around 40% of oligosaccharide chains were sialylated in F241A and V264A mutants relative to wild-type IgG3, which contained negligible levels (4% sialylation) prior to the mutations (Jefferis & Lund, 1997; Lund et al., 1996). The individual D265A and R301A mutants increased IgG3 sialylation to a lesser extent of about 30% and 16%, respectively (Jefferis & Lund, 1997; Lund et al., 1996). Crystallographic analysis and nuclear magnetic resonance (NMR) analysis revealed that the F241A and F243A replacement caused localized destabilization of the protein-glycan interface in the CH2 domain, indicating a greater degree of conformational flexibility and mobility compared to the native Fc (Jefferis & Lund, 1997; Subedi & Barb, Yu et al., 2013), enabling the carbohydrate to obtain increased access to glycosyltransferases (Mimura et al., 2016).

Taking together, the combination of IgG protein mutations with sialyltransferase overexpression can further maximize the sialylation content on antibody Fc region. In one study, the F243A replacement resulted in 53% α 2,3 sialylation of either one or both branches of the N-glycans of a recombinant IgG3 produced in CHO, and following transfection of rat ST6GAL1 into this line, 60% of IgG-Fc-linked glycans were S1 or S2 sialylated with a mixture of α 2,3 and α 2,6 sialic acids (Jassal et al., 2001). Our group investigated single and multiple amino acid substitutions of the Fc region with sialyltransferases knock-in and knock-out strategy to alter the sialylation pattern on IgGs expressed from CHO cells (Chung, Wang, Yang, Yin, et al., 2017). Four single amino acid mutations (F241A, F243A, V262E, and V264E) as well as combining all mutations in IgG expressing CHO cells increased α 2,3 sialylation and combining all the 4 mutations resulted in the highest levels of α 2,3 sialylation in unmodified CHO-K1 cells (Chung, Wang, Yang, Yin, et al., 2017). Additionally, engineering CHO cells containing dual knockouts of the ST3GAL4 and ST3GAL6 encoding

α 2,3 sialyltransferase together with overexpression of α 2,6 sialyltransferase (ST6Gal1) resulted in the exclusively α 2,6 sialylated antibodies (Chung, Wang, Yang, Yin, et al., 2017). One stable clone isolated from IgG1 4 amino acid substitutions in the α 2,3 sialyltransferase knockout- α 2,6 sialyltransferase knockin cells contained over 60% disialylated glycans (G2FS2) and approximately 80% glycans sialylated on one or both branches (Chung, Wang, Yang, Yin, et al., 2017). These studies demonstrated a new strategy in antibody glycan design incorporating genome and protein engineering in order to control and tailor specific glycan profiles.

2.2.2 | Antibody Fc sialylation may affect ADCC and CDC activities

Distinct from the anti-inflammatory activity, the sialylation of IgG Fc region can also affect other properties of antibody, such as ADCC and CDC activities. So far, the influence of antibody sialylation on ADCC activity has yet been settled and varying findings exist. For example, expression of exogenous sialidase A gene in mammalian cells resulted in the expression of a soluble enzyme that removed sialic acid from antibodies secreted into the medium. ADCC assay showed that the two IgG1 antibodies expressed in sialidase A-transfected cells had greater than 10-folds improved potency compared to the control mock-transfected IgG1s (Naso et al., 2010). The removal of terminal sialylation also increased the ADCC activity of pertuzumab (Luo et al., 2017). Alternatively, several recent investigations indicated that Fc sialylation does not impact the *in vitro* Fc γ RI and Fc γ RIIIa receptor binding and ADCC activity of IgG1, whether it is mono- or di-sialylated glycans (Quast et al., 2015; Thomann et al., 2015). However, mono-sialylated IgG1 showed a slight binding improvement to Fc γ RIIIa than bi-sialylated IgG1 (Boyd, Lines, & Patel, 1995; Thomann et al., 2015). Interestingly, the elimination of core fucosylation and its impact on Fc γ RIIIa binding affinity and ADCC activity is independent of terminal sialylation (Li et al. 2017; Luo et al. 2017). Regarding CDC activity, the presence of terminal sialic acid was found to lower IgG1 C1q binding and reduce C3b deposition on the cell surface (Quast et al., 2015). While antibody sialylation has been shown to promote anti-inflammatory activity in a α 2,6 sialylation manner, the mechanisms of antibody sialylation affecting other effector functions, such as ADCC, has rarely been studied and further exploration is needed to elucidate the functions of Fc sialylation.

2.3 | Galactosylation

In addition to low sialylation content of Fc glycans on both human serum IgG and recombinant IgGs, the Fc glycans can contain 0, 1, or 2 terminal Gal residues (G0, G1, or G2) with and without fucose on their antennae (Raju, 2008). Recombinant IgGs expressed in CHO cells are often undergalactosylated and tend to contain higher levels of G0 or G0F glycans compared with IgGs produced in mouse myeloma cell lines (Raju & Jordan, 2012). However, the predominant glycoforms on recombinant antibodies may vary depending on the cell lines, culture conditions, and substrates available in the media (Liu et al., 2014; Villacr s et al., 2015). The terminal galactosylation of IgG has been reported to affect CDC and ADCC (Hodoniczky, Zheng, & James, 2005; Peipp et al., 2008; Shinkawa

et al., 2003), but not impact antibody binding to the antigen (Raju & Jordan, 2012). Removal of terminal galactose residues from a chimeric mouse-human IgG1 antibody was shown to reduce CDC (Boyd et al., 1995). For Rituxan, an increase in terminal Gal content can increase CDC activity as a result of increased antibody binding to C1q (Hodoniczky et al., 2005). Similarly, the G1 glycoform of Rituximab triggered a CDC response twice as large as that triggered by the G0 glycoform (Jefferis, 2009). Nevertheless, the CDC improved efficacy is restricted to IgG1 and IgG3 subclasses, which suggests that Fc-galactosylation alone is not sufficient for IgG2 and IgG4 to promote CDC activity (Peschke et al., 2017). Interestingly, recent reports revealed that Fc galactosylation induces a positive impact on the binding affinity of IgG1 to FcγRIIIa and IIIa receptors and ADCC activity (Subedi & Barb, 2016; Thomann et al., 2015; Thomann, Reckermann, Reusch, Prasser, & Tejada, 2016). However, ADCC enhancement induced by Fc hypergalactosylation is not as robust as that caused by Fc afucosylation, which appears to provide a larger driving force for changing antibody ADCC activity (Thomann et al., 2016).

Overexpression of β1,4-galactosyltransferase in CHO cells has been demonstrated to enrich recombinant protein galactosylation and sialylation (Jeong et al., 2008; Raymond et al., 2015). Another concern when looking at the in vivo functional influence of terminal galactose is that sialic acid addition covers the penultimate galactose, as depicted in Figure 4 (Vidarsson, Dekkers, & Rispens, 2014). Thus, the enhancement of galactosylation is followed by an increase in sialylation content. IgG1 Abs lacking galactose and consequently sialic acid have shown aberrant functions that contributed to pro-inflammatory activity and inflammatory disorders (Axford et al., 1992; Rademacher et al., 1988). Besides, the combination of uridine, manganese chloride, and galactose can give a higher level of antibody galactosylation than individual additions, and the exact modulation of antibody galactosylation can be achieved by changing their concentrations in the culture media (Gramer et al., 2011).

Recently, our group observed that disruption of two α2,3 sialyltransferase (ST3GAL4 and ST4GAL6) from CHO cells expressing a mutant IgG with a mutated F241A Fc region generated an IgG containing 80% bigalactosylated and fucosylated (G2F) glycoforms (Chung, Wang, Yang, Ponce, et al., 2017). Expression of the F241A mutant IgG in triple gene knockout (FUT8^{-/-}, ST3Gal4^{-/-}, and ST3Gal6^{-/-}) CHO cells lowered the galactosylation glycoprofile to 65% bigalactosylated G2 glycans (Chung, Wang, Yang, Ponce, et al., 2017). However, expression of IgG with four amino acid substitutions (F241A, F243A, V262E, and V264E) in that triple gene knockout cells brought the G2 glycoform composition back up to approximately 75% (Chung, Wang, Yang, Ponce, et al., 2017). Thus, combining genome and protein engineering can be used to manipulate the galactosylation content on recombinant antibodies in CHO cells.

2.4 | GnT-I knockout to generate high-mannose glycans

Aside from complex and hybrid N-glycan types, high mannose glycans (Man₅₋₉GlcNAc₂) are characterized by five to nine terminal mannose residues attached to the GlcNAc₂ core (Kang et al., 2015). These

glycoforms are not widely observed, often accounting for less than 5% of the glycoprofile on many recombinant therapeutic antibodies, but they can have a major impact on in vivo therapeutic activity—enhancing ADCC, decreasing CDC, and increasing clearance rate (Reusch & Tejada, 2015; Wright & Morrison, 1994; Yu et al., 2012; Zhong et al., 2012). Also, high-mannose glycans have been widely applied to generate minimally glycosylated glycoproteins for X-ray crystallography (Chen & Stanley, 2003).

Golgi N-acetylglucosaminyltransferase I (GnT-I, also called Mgat1) is a type II transmembrane enzyme that plays a key role in the biosynthesis of N-linked glycans (Stanley, 2002). As shown in Figure 4, it specifically transfers GlcNAc from UDP-GlcNAc onto the α1,3 branch of the oligomannose core to initiate hybrid and subsequently complex N-linked carbohydrate synthesis in the medial Golgi (Stanley, 2002). Thus, methods for increasing high mannose levels focus on inhibiting glycosylation in or before the medial-Golgi, mainly by knocking down or knocking out the GnT-I gene.

For high mannose glycans, one of the most widely used and numerous studied cell lines is the Lec1 CHO mutant cell line (Shi & Goudar, 2014). Owing to a defect in GnT-I, Lec1 cells synthesize only oligomannose-type N-glycans (Stanley, 1983, 2002; Stanley & Chaney, 1985). High mannose oligosaccharide can vary from Man5 to Man9 in Lec1 cells (Kanda, Yamada et al., 2007; Wright et al., 2000; Zhong et al., 2012), and these cells have had limited applications in industry. Recently, a new high mannose cell line alternative to Lec1 has been established using genetic techniques in order to study the role of GnT-I in glycosylation. Sealover et al. (2013) reported the use of zinc-finger nuclease (ZFN) genome editing technology to disrupt GnT-I gene in CHO, generating cell lines that produce recombinant proteins predominantly bearing Man5 glycans. They successfully generated five clones out of 90 (5.6%) containing deletions in the GnT-I gene without lectin enrichment, and all five GnT-I-deficient cell lines demonstrated similar growth and productivity characteristics to the wild-type CHO-K1 host cell line (Sealover et al., 2013). Alternatively, Lee et al. (2015) demonstrated efficient targeted gene integration into site-specific loci in CHO through CRISPR/Cas9 genome editing together with a compatible donor plasmid harboring short-homology arms, the gene of interest (GOI), and a fluorescent marker gene as an indicator of random integration. By applying CRISPR/Cas9 and homology-directed DNA repair (HDR) mediated targeted integration, the GnT-I locus was completely disrupted in most target clonal cells (20 out of 25 clones), which showed a relative GnT-I copy number close to zero compared with that of 1 for GnT-I in CHO-S wild-type cells (Lee, Kallehauge, Pedersen, & Kildegaard, 2015).

2.5 | Obtaining homogeneous glycan patterns

Another characteristic of N-glycans for both native glycoproteins and recombinant antibodies is the presence of heterogeneous glycoforms, which can present challenges in protein purification, product characterization, and consistency and lot-to-lot reproducibility, potentially altering therapeutic efficacy. For example, this diversity can adversely affect drug potency and pharmacokinetics (Elliott et al., 2004; Ferrara, Brunner et al., 2006). Heterogeneity is attributed to the

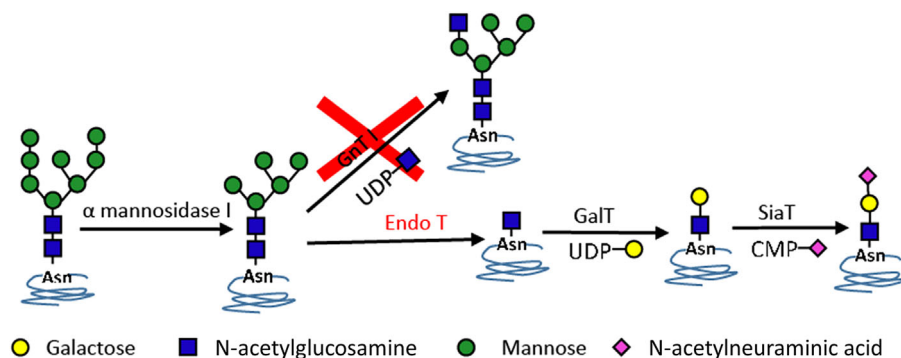


FIGURE 6 Schematic of GlycoDelete strategy, modified from Meuris et al. (2014)

lack of 100% enzymatic efficacy for each step in mammalian N-glycan biosynthesis and thus incomplete modification of glycans along the processing pathway (Meuris et al., 2014).

To solve this issue, Meuris et al. (2014) introduced a new strategy (GlycoDelete) in order to shortcut N-glycosylation by producing simple and homogenous glycoproteins expressed in HEK293S cells, as shown in Figure 6. Briefly, the authors inactivated N-acetylglucosaminyl-

transferase I (GnT-I) to produce high-mannose glycans. Then they overexpressed an endo-N-acetylglucosaminidase (Endo T) targeted to the trans-Golgi, followed by *Concanavalin A* (ConA) lectin selection. With the follow-on activities of galactosyltransferases (GalT) and sialyltransferases (SiaT) in the Golgi, GlycoDelete cells can produce proteins with the Gal-GlcNAc disaccharide or its α 2,3-sialylated trisaccharide derivative (Meuris et al., 2014). To evaluate the potential

TABLE 1 Summary of defucosylation strategies in mammalian cells

Genetic engineering			
Enzyme name	Enzyme function	Approach	Result
α -1,6 fucosyltransferase (FUT8) (-)	Transfer fucose to the innermost GlcNAc of the tri-mannosyl core structure	siRNA	Reduced FUT8 mRNA expression and antibody fucosylation
		Homologous recombination, ZFN,TALEN, CRISPR	Completely eliminate core fucosylation on recombinant antibodies in CHO
β -1,4-N-acetylglucosaminyltransferase III (GnT-III) (+)	Generate bisecting glycans	Overexpression of GnT-III; Co-expressing both GnT-III and Man II	Increased bisected non-fucosylated glycans and resulted in enhanced ADCC activity
GDP-fucose transporter (GFT) (-)	Transfer GDP-fucose into the Golgi apparatus	siRNA	Reduce Golgi GDP-fucose concentration, GFT mRNA expression and IgG fucosylation
GDP-mannose 4,6-dehydratase (GMD). (-)	Convert GDP-mannose to GDP-4-keto-6-deoxymannose	Co-knockdown of FUT8 and GMD using siRNA tandem	Convert antibody-producing CHO cells to fully non-fucosylated antibody producers in serum-free fed-batch culture
prokaryotic GDP-6-deoxy-D-lyxo-4-hexulose reductase (RMD) enzyme (+)	Convert GDP-mannose to GDP-6-deoxy-d-lyxo-4-hexulose	overexpression	Reduces the GDP-6-keto-6-deoxy-mannose synthesis in the de novo pathway and inhibit fucosylation
GDP-4-keto-6-deoxymannose-3,5-epimerase-4-reductase (GMER, or FX protein) (-)	Convert GDP-4-keto-6-deoxymannose to GDP-fucose in the de novo fucosylation pathway	CRISPR	Express antibodies with comparable attributes as parental host with a desired ratio of primarily fucosylated to afucosylated glycans when L-fucose is titrated into the production media
Media supplementation			
Chemical name	Mechanism	Result	
D-arabinose	A chemical analog of L-fucose	Reduce high-mannose N-glycans and provide nearly 100% replacement of fucosylation with arabinosylation	
2-deoxy-2-fluoro-L-fucose	Fucosylation inhibitor	Reduce antibody fucosylation	

Elimination or inhibition of fucosylation can increase Fc binding to Fc γ RIII receptor and remarkably increase antibody ADCC activity. (-): Knockdown or knockout; (+): Knockin or overexpression.

of these GlycoDelete glycans and their influence on therapeutic proteins, human GM-CSF (hGM-CSF) and an anti-CD20 antibody were produced in HEK293s and HEK293sGlycoDelete cells (Meuris et al., 2014). For hGM-CSF the presence of altered glycans obtained using the GlycoDelete cells had no detectable effect on the activity of the protein generated by these cells. On the other hand, GlycoDelete anti-CD20 exhibited a significantly reduced Fc γ RIII affinity and an increased circulation time in mice compared to HEK293s produced anti-CD20 (Lepenies & Seeberger, 2014; Meuris et al., 2014).

Finally, in order to provide homogenous glycoforms, Zhang et al. (2010) conducted a comprehensive Zinc-finger nuclease knockout screen of 19 glycosyltransferase genes and identified the key genes that control decisive steps in N-glycosylation in CHO (Yang et al., 2015). The authors stacked knockouts of GnT-IV-A/GnT-IV-B/GnT-V to produce almost homogenous biantennary N-glycans (Yang et al., 2015). Subsequently, the introduction of ST6Gal-I in CHO ST3Gal4/6 knockout cells produced a normal range of N-glycans with only α 2,6-sialylation on recombinant IgG, and when combined with a GnT-IV-A/GnT-IV-B/GnT-V knockout, homogenous biantennary N-glycans capped by α 2,6-sialylation were generated (Yang et al., 2015).

3 | CONCLUSIONS

The effector functions of monoclonal antibody are highly dependent on the structure of the N-linked glycans in the Fc domain of the antibody and numerous investigations over past decades have evaluated methodologies to control the Fc glycan structures. These glycoforms include core fucosylation (summarized in Table 1), terminal galactosylation and sialylation, high-mannose, and bisecting glycans via multiple strategies. There are a number of pros and cons for each of the many approaches discussed in this review. Cell engineering approaches are typically performed most effectively at an early stage of cell line generation. Its implementation later in the process development may not be feasible. Media modification is a simpler approach to change the glycoprofile but the resulting glycans may not be precisely controlled. Furthermore, the alteration of one specific glycan form may affect other glycan forms, such as increasing bisecting glycans will inhibit core fucosylation and increasing galactosylation will commitantly improve sialylation. The examination and integration of multiple complementary approaches can help to optimize and advance glycan profiles that cannot be achieved by a single approach. For example, multiple glycogene knock-outs and knock-ins can generate exclusively α 2,6 sialylated antibodies. Another example involves blocking the de novo pathway by knocking out the FX/GMER gene combined with L-fucose feeding in order to fine tune the antibody fucosylation pattern without affecting other quality attributes. This review has detailed a number of different cell engineering and media modification approaches to tailor the glycans on antibodies. Choosing which approaches to apply depends on the specific purpose, the development stage and efficiency of the technique or techniques as well as other factors including biomanufacturing compatibility, relative ease of implementation, and intellectual patent considerations among others.

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CONFLICTS OF INTEREST

The authors declare that there is no conflict of interest regarding the publication of this article.

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